

Bio-ORACLE v2.0: Extending marine data layers for bioclimatic modelling

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Abstract

Motivation: The availability of user-friendly, high-resolution global environmental datasets is crucial for bioclimatic modelling. For terrestrial environments, WorldClim has served this purpose since 2005, but equivalent marine data only became available in 2012, with pioneer initiatives like Bio-ORACLE providing data layers for several ecologically relevant variables. Currently, the available marine data packages have not yet been updated to the most recent Intergovernmental Panel on Climate Change (IPCC) predictions nor to present times, and are mostly restricted to the top surface layer of the oceans, precluding the modelling of a large fraction of the benthic diversity that inhabits deeper habitats. To address this gap, we present a significant update of Bio-ORACLE for new future climate scenarios, present-day conditions and benthic layers (near sea bottom). The reliability of data layers was assessed using a cross-validation framework against in situ quality-controlled data. This test showed a generally good agreement between our data layers and the global climatic patterns. We also provide a package of functions in the R software environment (*sdmpredictors*) to facilitate listing, extraction and management of data layers and allow easy integration with the available pipelines for bioclimatic modelling.

Main types of variable contained: Surface and benthic layers for water temperature, salinity, nutrients, chlorophyll, sea ice, current velocity, phytoplankton, primary productivity, iron and light at bottom.

Spatial location and grain: Global at 5 arcmin (c. 0.08° or 9.2 km at the equator).

Time period and grain: Present (2000–2014) and future (2040–2050 and 2090–2100) environmental conditions based on monthly averages.

Major taxa and level of measurement: Marine biodiversity associated with sea surface and epibenthic habitats.

Software format: ASCII and TIFF grid formats for geographical information systems and a package of functions developed for R software.

KEYWORDS

Bio-ORACLE, bioclimatic modelling, environmental data, global, kriging, macroecology, marine, species distribution modelling

1 | INTRODUCTION

Early attempts to model the relationship between the occurrence or abundance of species and their natural environment relied heavily on

environmental variables measured in situ (Sutherland & Maywald, 1985) and often involved complex software pipelines specifically developed to extract, organize and visualize data (Kemp, Loon, Shamoun-Baranes, & Bouten, 2012; Lima-Ribeiro et al., 2015). The spatial and temporal

resolution of environmental data also showed high variability, precluding smooth integration and comparison of bioclimatic analyses (Lima-Ribeiro et al., 2015). Alongside the development of geographical information systems (GIS), the advent of cutting-edge spatial interpolation resulted in data layers representing global environmental conditions, conformal in extent and resolution. Pioneer initiatives, such as the Climatic Research Unit Terrestrial Climatology (New, Hulme, & Jones, 1999) and WorldClim (Hijmans, Cameron, Parra, Jones, & Jarvis, 2005a), significantly pushed the application of bioclimatic modelling in ecology, biogeography, conservation biology and evolution. Yet, these gridded datasets were tailored for terrestrial climates only, and the availability of marine data layers lagged significantly behind (Robinson et al., 2011). National Oceanic and Atmospheric Administration's World Ocean Atlas (Levitus, 2001), AquaMaps (Kaschner et al., 2008) and Hexacoral (Fautin & Buddemeier, 2008) were only recently enhanced by the more comprehensive and higher resolution datasets Bio-ORACLE (Tyberghein et al., 2012) and Marspec (Sbrocco & Barber, 2013).

The increased accessibility of marine data layers allowed an emerging body of research to describe the global distribution of species (e.g., Chaudhary, Saeedi, & Costello, 2017; Chefaoui, Assis, Duarte, & Serrão, 2015; Hill & Terblanche, 2014; Parravicini et al., 2013; Stuart-Smith, Edgar, Barrett, Kininmonth, & Bates, 2015), address niche-based questions (e.g., Assis et al., 2015; Lee-Yaw et al., 2016; Verbruggen et al., 2009), support biodiversity conservation (e.g., Boavida, Assis, Silva, & Serrão, 2016; Guisan & Thuiller, 2005; Guisan et al., 2013) and ecosystem-based management (Hobday, Hartog, Timmiss, & Fielding, 2010) and infer the likely anthropogenic pressures leading to population turnover and extinction (e.g., Scherner et al., 2013). The establishment of standard protocols (e.g., Coupled Model Intercomparison Project; CMIP) delivering the outputs of atmosphere-ocean general circulation models (AOGCMs) for past and future climate scenarios (Moss et al., 2010; Otto-Bliesner et al., 2009) further expanded the applications for marine data layers, for instance, to predict range shifts through time (e.g., Assis, Bercibar et al., 2017; Burrows et al., 2014; Neiva et al., 2015; Thomas et al., 2004) or test relevant evolutionary hypotheses such as the location of marine biodiversity hotspots free from past bottlenecks and extinctions (i.e., climatic refugia; Assis, Serrão, Claro, Perrin, & Pearson, 2014; Waltari et al., 2007).

The marine datasets currently available, however, are almost exclusively restricted to the top surface layer of the oceans (e.g., Bio-ORACLE), and those including benthic layers adjacent to the seabed are particularly coarse in resolution (Hexacoral to c. 56 km² and World Ocean Atlas to c. 112 km², at the equator) or limited to biophysical features extracted from bathymetric profiles (MARSPEC; Sbrocco & Barber, 2013). These constraints significantly limit the potential for modelling benthic species (Boavida et al., 2016; Davies & Guinotte, 2011; Reiss et al., 2014), which include a large proportion of marine biodiversity. For instance, the exploration of deep cryptic refugia for marine species is suboptimal when using surface data only (Assis et al., 2016; Graham, Kinlan, Druehl, Garske, & Banks, 2007; Perry, Low, Ellis, & Reynolds, 2005).

To address this gap, we present a significant extension of the marine data layers available in Bio-ORACLE. New ecologically relevant surface and benthic layers tailored for mechanistic and correlative modelling (Kearney & Porter, 2009; Peterson et al., 2011) are provided for present conditions and the new generation of climate change scenarios (Moss et al., 2010). Besides the extension of Bio-ORACLE to include benthic layers for temperature, salinity, nutrients and chlorophyll (Table 1), we also provide new data on sea ice, current velocity, phytoplankton, primary productivity, iron and light at the bottom for a better understanding of marine macroecological processes. We also determine the reliability of data layers (as stressed by Hall & Hall, 2014) using a cross-validation framework against in situ quality-controlled data. We provide a package of functions in the R software environment (R Development Core Team, 2016) for easy integration with the available pipelines for bioclimatic modelling (e.g., Naimi & Araújo, 2016; Thuiller, Lafourcade, Engler, & Araújo, 2009).

2 | MARINE DATA LAYERS

Marine data layers for present conditions were produced with climate data describing monthly averages for the period 2000–2014, obtained from pre-processed global ocean re-analyses combining satellite and in situ observations at regular two- and three-dimensional spatial grids (Table 1). Future layers were produced for 2040–2050 and 2090–2100 by averaging data from distinct AOGCMs provided by the CMIP 5 (Table 2). The available data (temperature, salinity, current velocity and sea ice thickness) were obtained for the new representative concentration pathway scenarios (RCP): the RCP26, a peak-and-decline scenario ending in very low greenhouse gas concentration levels by the end of the 21st century; the RCP45 and RCP60, in which levels stabilize; and the RCP85, a scenario of increasing emissions over time, leading to high greenhouse gas concentration levels (reviewed by Moss et al., 2010).

The monthly averages for the present and future were used to produce six distinct predictors per variable for bioclimatic modelling: the long-term average; the minimum and maximum records; the long-term average of the minimum and maximum records per year (e.g., temperature of the warmest month, on average); and range, given by the average of the absolute difference between the minimum and maximum records per year. These predictors were statistically downscaled (i.e., from coarse- to fine-scale resolution) to a common spatial resolution of 5 arcmin (c. 0.08° or 9.2 km at the equator) by fitting a kriging model based on the 12 nearest values of each focal cell (e.g., Hofstra, Haylock, New, Jones, & Frei, 2008; Lima-Ribeiro et al., 2015). The choice of kriging over other interpolation methods was based on studies showing higher performance for this method (e.g., Hofstra et al., 2008; Lima-Ribeiro et al., 2015) and also on a priori testing performed against inverse distance weighting (IDW; e.g., Assis et al., 2014; Kemp et al., 2012). This test cross-validated the interpolation of 1×10^4 random records with both methods for different variables, and showed lower root mean square error (RMSE) and mean absolute error (MAE) for kriging, despite the lack of differences in the mean value of all

TABLE 1 Marine data layers, units, correspondence with the first version of Bio-ORACLE, range of values (determined for benthic layers, at their average depth), accuracy assessed with quality-control data, number of quality-control records (*n*) and source of climate data

Layer	Units	BO1	Range	MAE	RMSE	Cor	<i>n</i>	Source
Temperature	°C	Yes	[−1.94;39.22]	0.39	0.75	0.99	445,248	ARMOR
Salinity	PSS	Yes	[4.75;41.96]	0.13	0.52	0.93	444,925	ARMOR
Sea ice concentration	Fraction	No	[0;1]	–	–	–	–	ORAP
Sea ice thickness	m	No	[0;10.94]	–	–	–	–	ORAP
Current velocity	$\text{m} \cdot \text{s}^{-1}$	No	[0;2.42]	–	–	–	–	ORAP
Nitrate	$\text{mmol} \cdot \text{m}^{-3}$	Yes	[0;164.51]	1.62	2.47	0.98	93,201	PISCES
Phosphate	$\text{mmol} \cdot \text{m}^{-3}$	Yes	[0;3.55]	0.15	0.23	0.97	349,074	PISCES
Silicate	$\text{mmol} \cdot \text{m}^{-3}$	Yes	[0.46;316.67]	5.98	9.05	0.99	367,629	PISCES
Dissolved molecular oxygen	$\text{mmol} \cdot \text{m}^{-3}$	Yes	[0;789.94]	15.27	23.50	0.96	417,790	PISCES
Dissolved iron	$\text{mmol} \cdot \text{m}^{-3}$	No	[0;0.03]	–	–	–	–	PISCES
Chlorophyll	$\text{mg} \cdot \text{m}^{-3}$	Yes	[0;17.46]	0.1	0.15	0.56	5,401	PISCES
Phytoplankton*	$\text{mmol} \cdot \text{m}^{-3}$	No	[0;44.94]	–	–	–	–	PISCES
Primary productivity*	$\text{g} \cdot \text{m}^{-3} \cdot \text{day}^{-1}$	No	[0;0.95]	–	–	–	–	PISCES
Light at the bottom	$\text{E} \cdot \text{m}^{-2} \cdot \text{yr}^{-1}$	No	[0;69.21]	–	–	–	–	GlobColour

ARMOR = Global Observed Ocean Physics Reprocessing (resolution: 0.25°/33 vertical levels); BO1 = first version of Bio-ORACLE; Cor = Pearson's correlation coefficient; GlobColour merging MERIS/MODIS/SeaWiFS (resolution: 0.05°/surface); MAE = mean average error; ORAP = Global Ocean Physics Reanalysis ECMWF (resolution: 0.25°/75 vertical levels); PISCES = Global Ocean Biogeochemistry Non-assimilative Hindcast (resolution: 0.25°/75 vertical levels); RMSE = root mean square error. *Expressed as carbon in sea water

[Correction added on 8 January 2018, after first online publication: Units have been corrected in this version]

variables (nonparametric Kruskal–Wallis *p*-values > .05; see Supporting Information Table S1.1).

The downscaling process for benthic layers considered the geographical position and depth of cells (e.g., Assis et al., 2016; Boavida et al., 2016) as inferred from the general bathymetric chart of the oceans (GEBCO, 2015). Given that focal cells included a range of depth values, the benthic layers were produced for the minimum, average and maximum depths. The future layers were downscaled using the change-factor approach (Lima-Ribeiro et al., 2015; Varela, Lima-Ribeiro, & Terribile, 2015; Wilby et al., 2004). This technique is based on applying the predicted magnitude of climate change to the data layers produced for the present. For this purpose, data for the period 2000–2014 were also obtained from the AOGCMs to determine the difference (change-factor) between the present conditions and the future scenarios of change, at the native resolution of each AOGCM (Table 2).

Next, the change-factor was downscaled to 0.08° resolution with kriging (as previously described) and applied to the corresponding base-line layer for the present conditions.

The layers for current velocity and light at the bottom were further post-processed. The current velocity was determined with the Pythagoras theorem on the meridional (along the longitude circle) and zonal (along the latitude circle) components of ocean currents, whereas light at the bottom used a standard depth-dependent exponential function (Assis et al., 2016; Graham et al., 2007) based on photosynthetically active radiation (PAR) and diffuse attenuation coefficient (Kd490):

$$\text{Light at bottom} = \text{PAR} \times \exp(-\text{Kd}490 \times z),$$

where *z* is depth inferred from the general bathymetric chart of the oceans (GEBCO, 2015).

TABLE 2 Average and range of climatic anomalies for the future (2090–2100) under different scenarios of change, and source of data (coupled atmosphere–ocean general circulation models)

Layer	RCP26	RCP45	RCP60	RCP85	Source (AOGCM)
Temperature	0.21 [−2.06;4.45]	0.29 [−2.01;6.72]	0.33 [−1.24;6.89]	0.51 [−2.62;10.98]	CCSM4, HadGEM2-ES, MIROC5
Salinity	−0.03 [−6.03;1.37]	−0.04 [−4.97;1.39]	−0.05 [−4.48;1.34]	−0.07 [−4.33;1.83]	CCSM4, HadGEM2-ES, MIROC5
Current velocity	0.01 [−0.17;0.18]	0.01 [−0.11;0.28]	0.01 [−0.13;0.16]	0.01 [−0.18;0.24]	CCSM4, HadGEM2-ES, MIROC5
Sea ice thickness	−0.12 [−8.33;1.43]	−0.18 [−8.78;1.98]	−0.22 [−8.95;1.54]	−0.27 [−9.01;1.88]	CCSM4, HadGEM2-ES, MIROC5

Note. Climatic anomalies were inferred by determining the difference between the benthic layers (average depth) produced for the different representative concentration pathway (RCP) scenarios and those for the present. CCSM4 = The Community Climate System Model 4 (resolution: 1.13° × 0.47°/60 vertical levels); HadGEM2-ES = Hadley Centre Global Environmental Model 2 (resolution: 1.00° × 0.84°/40 vertical levels); MIROC5 = Model for Interdisciplinary Research on Climate 5 (resolution: 1.41° × 0.81°/50 vertical levels).

All layers were exported to ASCII (ESRI) and TIFF grid formats for easy downloading and integration in modern GIS technologies (e.g., ESRI and QGIS). To minimize the possible spatial gap with 'no data' between the data layers provided and the available vector shorelines, the global self-consistent, hierarchical, high-resolution geography database (Wessel & Smith, 1996) is recommended.

3 | RELIABILITY OF MARINE DATA LAYERS

The reliability of downscaled data layers was inferred with in situ quality-controlled data provided by the Global Ocean Data Analysis Project (GLODAP; Olsen et al., 2016) for temperature, salinity, phosphate, nitrate, silicate, dissolved oxygen and chlorophyll. This was performed by cross-validating the outputs of downscaling the raw data used to produce data layers at the locations (geographical position and depth) of each sample available in GLODAP, with the actual data provided by this dataset (e.g., Assis, Bercibar et al., 2017; Boavida et al., 2016; Davies & Guinotte, 2011). The paired relationships between the interpolated and in situ data were statistically analysed with mean absolute error, root mean square error and Pearson's correlation. These tests showed the layers mirroring most climatic patterns present in quality-controlled data. All variables retrieved low error rates (Table 1) and high correlation coefficients (Pearson's correlation: $>.93$; Table 1) with a unique exception for chlorophyll, which showed a much lower correlation (Pearson's correlation: $.56$; Table 1; see Supporting Information Figure S1.1). This exception is likely to result from coupling the high temporal variability known for chlorophyll on a daily basis (e.g., Iriarte, González, Liu, Rivas, & Valenzuela, 2007; Wang, Le Borgne, Murtugudde, Busalacchi, & Behrenfeld, 2009; Wang Hladik et al., 2010) with the monthly averages used to produce the layers (as discussed by Boavida et al., 2016; Davies & Guinotte, 2011). Moreover, chlorophyll was the variable with the lowest number of in situ observations (5,401 for the global

ocean; Table 1), a fact that might have precluded a proper assessment of reliability in cross-validation.

The spatial distribution error of the layers was illustrated by mapping the difference between the interpolated and in situ data onto a 2.5° grid (e.g., Davies & Guinotte, 2011) and by plotting this difference against depth. In general, the distribution of errors also showed high accuracy for all layers. Temperature, phosphate, nitrate and dissolved molecular oxygen displayed only specific anomalies, highly restricted to discrete regions of the global ocean (e.g., east Siberian Sea and southern North Sea; Supporting Information Figures S1.4, S1.8, S1.10 and S1.14), and with no relationship with depth (Supporting Information Figures S1.3, S1.7, S1.9 and S1.13). The errors for silicate were mostly in the Southern Ocean (Supporting Information Figure S1.12) and those for salinity were in the top layers (surface waters with higher positive anomaly; Supporting Information Figure S1.5) of the Canadian Arctic and east Siberian Sea (Supporting Information Figure S1.6), particularly for values below 30 PSS (Supporting Information Figure S1.5). The spatial distribution of errors also showed that dissolved molecular oxygen, phosphate, salinity, silicate and temperature have good coverage of in situ samples (GLODAP dataset), whereas nitrate and chlorophyll are mostly uncovered throughout the globe.

4 | R-PACKAGE TOOL

In addition to providing the layers for downloading in ASCII and TIFF formats, we also developed the *sdmpredictors* package of functions in R (R Development Core Team, 2016) to facilitate listing, extraction and management of data layers. This package, whose functions are detailed in Table 3, also integrates the layers from the first version of BioORACLE, as well as those of MARSPEC (Sbrocco & Barber, 2013) and BioClim (Hijmans, Cameron, Parra, Jones, & Jarvis, 2005b). The source code and related help files are available via the CRAN repository and

TABLE 3 List of functions available in *sdmpredictors* package, description and arguments that users may specify

Function	Description	Arguments
<code>list_datasets()</code>	Explore datasets available in the package	Terrestrial (logical) only terrestrial data are returned; marine (logical) only marine data are returned
<code>list_layers()</code>	Explore layers in a dataset	Datasets (character); terrestrial (logical); marine (logical); monthly (logical) only annual and monthly layers are returned (default)
<code>list_layers_future()</code>	Explore future layers in the package	Datasets (character); scenario (character) for climate change scenario (e.g., RCP85); year (integer) for the climate change prediction (e.g., 2100); terrestrial (logical); marine (logical); monthly (logical)
<code>get_future_layers()</code>	Get the name of a future climate layer(s) based on the current climate layer(s)	Current_layer_codes (character) with the code(s) of the layers either as a vector or dataframe provided by <code>list_layers()</code> ; scenario (character); year (integer)
<code>load_layers()</code>	Download specific layers to the current directory	Layercodes (character) with the codes of the layers to be loaded; equalarea (logical) for Behrmann cylindrical equal-area projection; rasterstack (logical) to stack the layers in a unique object; datadir (character) for the directory to store data
<code>layer_stats()</code>	Layer statistics	Layercodes (character)
<code>layers_correlation()</code>	Pearson's correlation coefficient between layers	Layercodes (character)

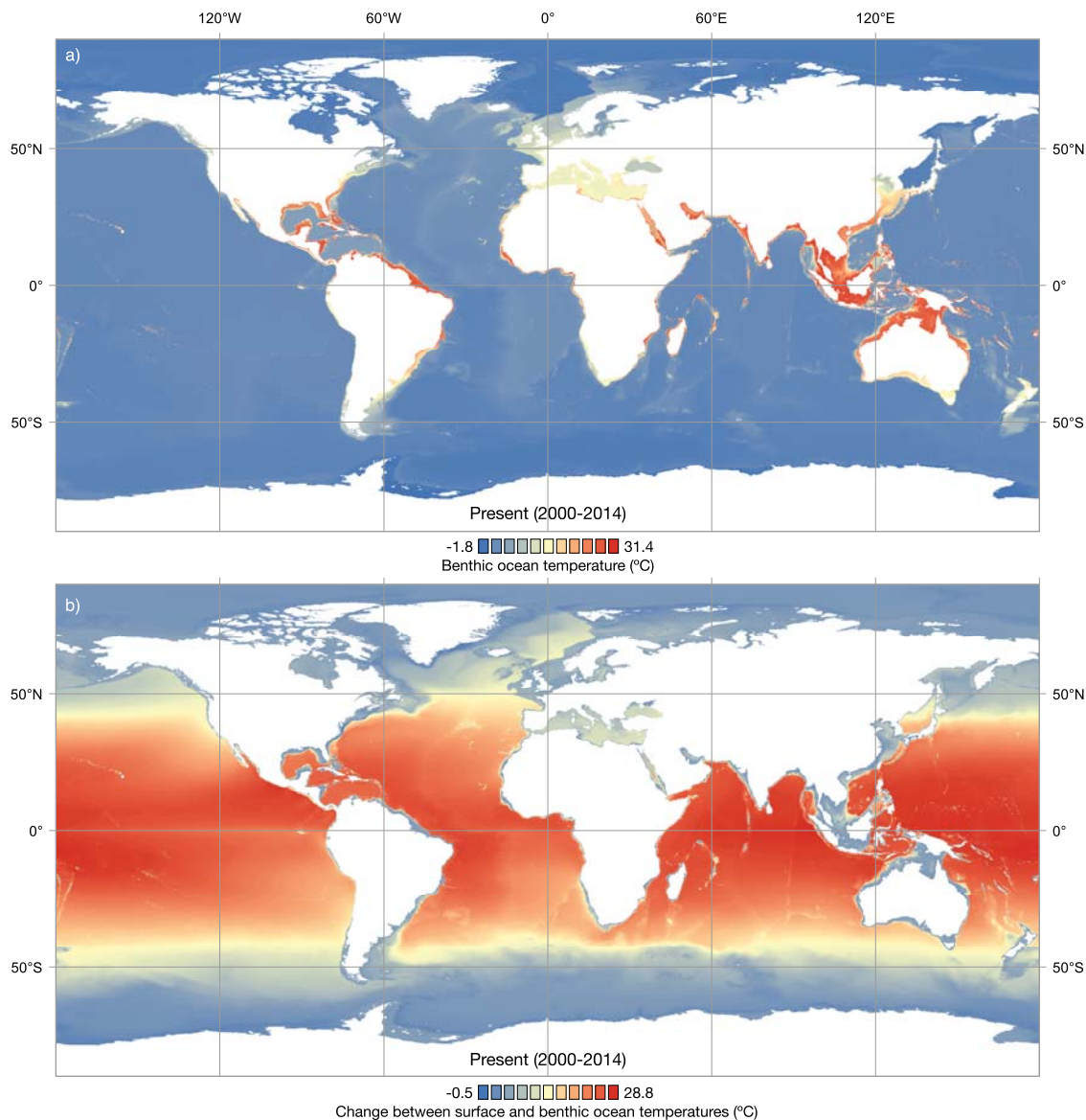


FIGURE 1 (a) Mean benthic ocean temperature for the present (period 2000–2014) and (b) change (difference) between surface and benthic ocean temperatures

can be installed easily by entering the following lines of code into the R command prompt:

1. `install.packages('sdmpredictors')`.
2. `library(sdmpredictors)`.

5 | CONCLUSION

A comprehensive set of new ecologically relevant data layers are provided for bioclimatic modelling in a user-friendly format, with global coverage and comparable grid system. The current update expands the potential of Bio-ORACLE by (a) covering the benthic realm with climate and environmental data, (b) adding new variables to present-day conditions (i.e., current velocity, iron, light, phytoplankton, primary productivity and sea ice), (c) adding the new generation of climate change

scenarios, (d) providing means of data reliability and uncertainty, and (e) providing a package of functions in the R software environment.

The relevance of new data aiming for species associated with sea benthic features (e.g., Assis et al., 2016; Boavida et al., 2016) is clearly underlined by the disparity in ocean temperatures between surface and benthic layers, which can amount to up to 28.8 °C in the deeper regions of lower latitudes (Figure 1). The data provided for the new generation of climate change scenarios further diversifies the range of scientific questions that can be addressed using Bio-ORACLE. One such case is the possibility to explore climate-induced depth range shifts (e.g., Assis et al., 2016; Assis, Araújo, & Serrão, 2017), the marine equivalent to elevation range shifts for terrestrial species (Chen, Hill, Ohlemüller, Roy, & Thomas, 2011; Galbreath, Hafner, & Zamudio, 2009). The reliability of new layers has been assessed with in situ quality-controlled data. This information is relevant for marine

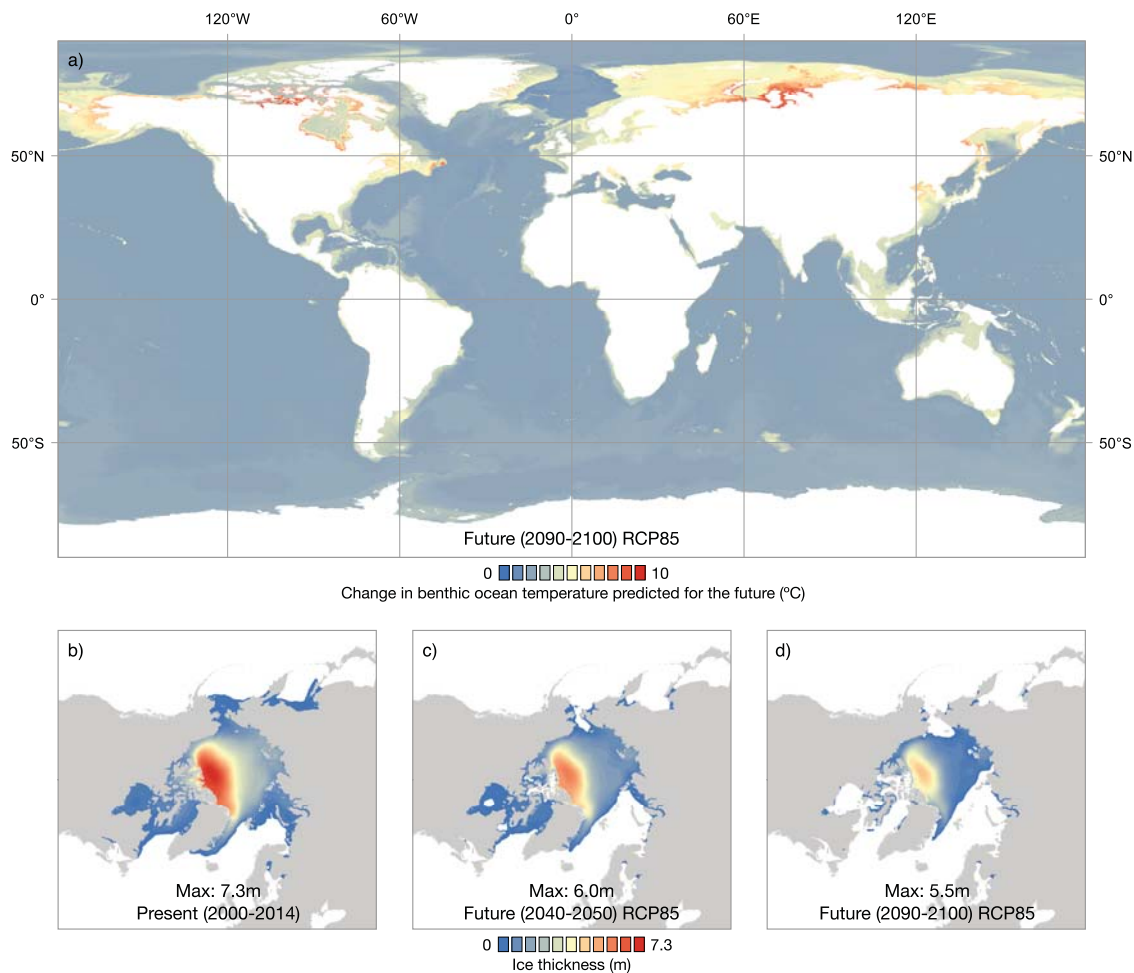


FIGURE 2 (a) Change in the maximal benthic ocean temperature predicted for the period 2090–2100 with RCP85. (b–d) Arctic mean ice thickness for the periods (b) 2000–2014, (c) 2040–2050 and (d) 2090–2100 predicted with RCP85

modellers and showed a good agreement between data layers and the global climatic patterns. The future data using the magnitude of climate changes (change-factor) of different AOGCMs provide the highest confidence level currently attainable (Hall & Hall, 2014). Despite their inherent uncertainties, the AOGCMs used represent the present scientific understanding linking greenhouse gas emissions with global climate changes.

The familiar data structure of Bio-ORACLE (rasters for GIS) and its integration with R computing language should allow easy acquisition, exploration and manipulation of data, as well as smooth integration with the available statistical tools (e.g., Naimi & Araújo, 2016; Thuiller et al., 2009). The new data layers represent a valuable addition to the spatial information about climate available for the global ocean. Their key features can be used to improve bioclimatic modelling and provide valuable insights into the current and future states of marine biodiversity and indirectly into the services it provides to society. When requested by decision-makers, these outcomes may guide important climate change-integrated conservation strategies (Hannah, Midgley, & Millar, 2002; Hobday et al., 2010) and feed baseline assessments, such as those required for the Intergovernmental

Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES).

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their model output. For CMIP, the U.S. Department of Energy's Program for Climate Model Diagnosis and Intercomparison provides coordinating support and led development of software infrastructure in partnership with the Global Organization for Earth System Science Portals.

DATA ACCESSIBILITY

The Bio-ORACLE layers are accessible online at <http://www.bio-oracle.org> for download (as ASCII and TIFF files) and preview.

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BIOSKETCH

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SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article .

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